COLM
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PSI-Bench: Interpretable and Clinically Meaningful Evaluation of Depression Patient Simulators

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Motivation

LLM-based simulator in counsellor training:

- Practical, scalable tool for sensitive training
- Should reflect realistic, diverse patient behaviors
- Simulating depression is challenging: Safety constraints, high variability across patients

Limitations of current automatic evaluations:

- Heavily rely on LLM-judges with underspecified rating prompts

  → Reported to contradict expert ratings; Lack interpretability
- Do not assess behavior diversity between different patients

Introducing PSI-Bench

Patient-Simulator-Bench (PSI-Bench): A framework for **interpretable**, **clinically meaningful** evaluation of depression simulators.

Interpretable:

- 5 evaluation dimensions: scores, analysis

Clinically meaningful

- Dimensions motivated by psycholinguistic findings on depression
- Compared against expert-curated depression-related conversations (Eeyore)
- Findings and rankings agree with counsellors judgements

PSI-Bench Overview

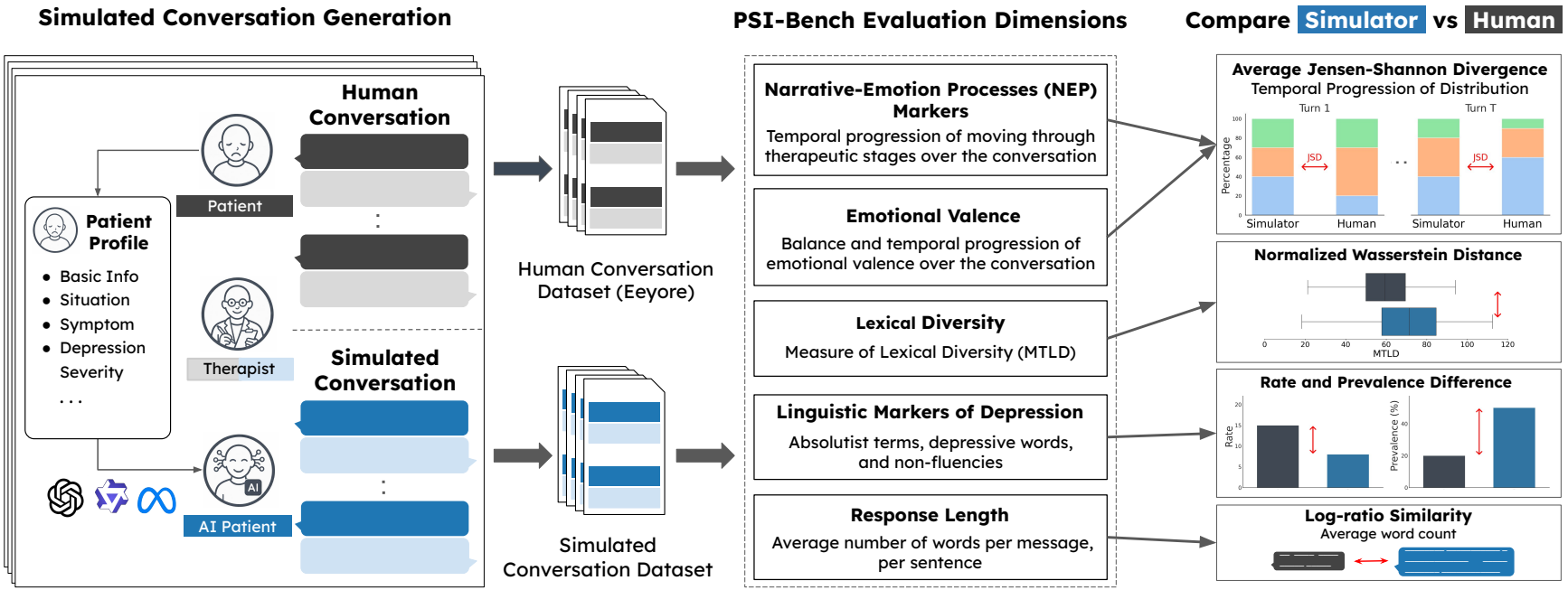
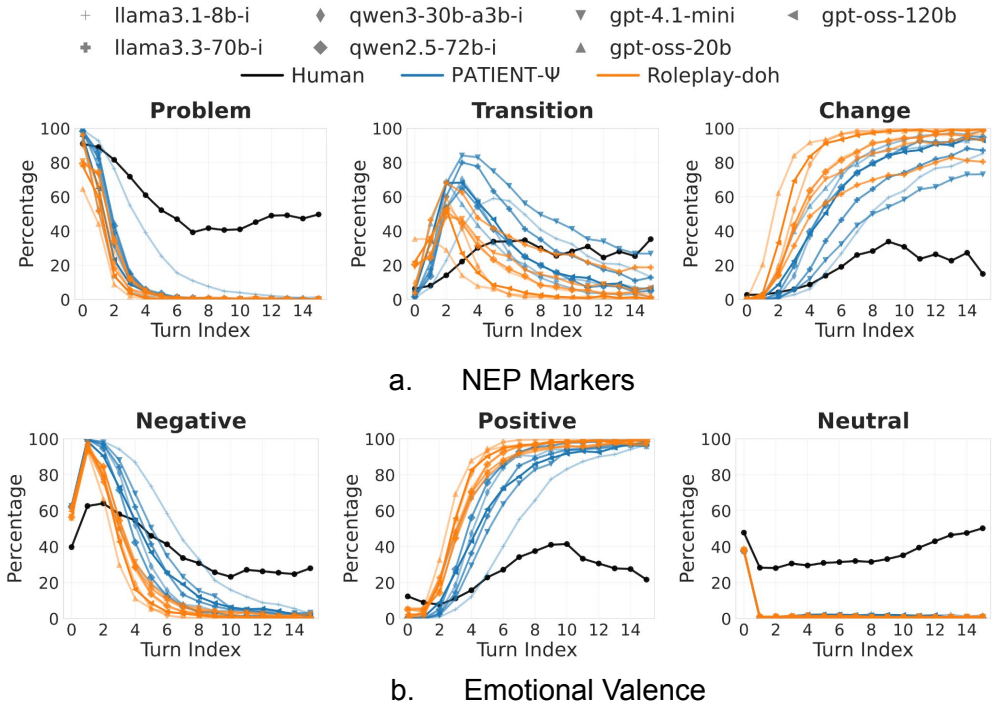


Fig 1. From depression-related human conversations (Eeyore), condition patient simulator on derived patient profiles to generate a simulated conversation dataset. Evaluate both datasets independently using PSI-Bench dimensions, and compare resulting statistics.

Experiment Setup: 14 simulation datasets: 7 LLMs x 2 methods (PATIENT-ψ, Roleplay-doh)

Results: RQ1 – Differences from Human



a. NEP Markers

b. Emotional Valence

Fig 2. Population-level progression of patient responses across turns. Each point represent % messages assigned to a category at a turn.

Progress through therapeutic stages faster than human patients

Follow a rapid negative-to-positive trajectory

(We analyze differences for all five dimensions)

Results: RQ2 – Impact of LLM, simulation method

- Simulation framework can matter more than LLM choice
- Larger models do not guarantee better realism

Rank	Setting	NEP ↑	Valence ↑	Div. ↑	Length ↑	Markers ↑	Overall ↑
1	PS-llama3.1-8b-i	**70.70**	**80.83**	72.17	28.24	57.05	61.80
2	PS-qwen3-30b-a3b-i	54.78	68.63	63.17	**43.89**	**67.97**	59.69
3	PS-qwen2.5-72b-i	56.78	67.95	57.25	42.88	61.76	57.33
4	RD-llama3.3-70b-i	57.01	66.98	72.86	29.41	52.35	55.72
5	PS-llama3.3-70b-i	58.93	73.04	70.57	24.18	51.37	55.62
6	RD-llama3.1-8b-i	52.37	65.27	**73.43**	25.24	44.33	52.13
7	RD-qwen2.5-72b-i	52.28	66.51	44.70	37.60	57.47	51.71
8	PS-gpt-4.1-mini	59.68	75.80	33.07	33.84	54.22	51.32
9	PS-gpt-oss-20b	52.80	68.30	43.74	28.76	52.59	49.24
10	RD-qwen3-30b-a3b-i	46.66	61.81	45.98	35.21	49.15	47.76
11	PS-gpt-oss-120b	55.40	74.17	22.66	26.00	54.97	46.64
12	RD-gpt-4.1-mini	54.24	69.29	18.98	30.95	53.37	45.36
13	RD-gpt-oss-20b	43.42	62.84	24.97	22.72	39.64	38.72
14	RD-gpt-oss-120b	46.11	63.52	0.00	20.77	36.14	33.31

Table 1. Benchmark results across all dimensions and overall score. Higher scores: higher alignment with human reference. PS–:PATIENT-Ψ, RD–:Roleplay-doh, Div.: lexical diversity. Rankings based on overall score (descending).

Results: RQ3 – Agreement with Counsellors

	Human–Model		Human–Human
Task	**% Agreement**	**Cohen's κ**	**Fleiss' κ**
Pairwise	91.67	0.8210	0.4328
NEP	80.06	0.6958	0.4997
Valence	85.33	0.725	0.5712

Table 2. **High agreement** with annotators' judgements for pairwise preference and LLM-based classifications.

Human Study: 20 mental health professionals

Qualitative Reasons:

- Less realistic: Overly polished, self-aware responses and progress fast
- Most realistic: came mostly from **top-ranked settings**: shorter, fragmented, hesitant

Thank you!

Any Questions?

Paper link: arxiv.org/abs/2604.25840

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